

Quantitative Assessment of Credit Portfolio Risk: An Empirical Study Based on the Single-Factor Gaussian Copula and Bernoulli Mixture Model

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Abstract

This report simulates and evaluates the one-year credit loss distribution of a given credit portfolio. To address the "messy data" typical of simulated real-world financial environments, this analysis first constructs a data cleaning pipeline using systematic mapping rules and mandatory validation, successfully merging portfolio holdings, firm characteristics, stock returns, and historical defaults. Based on this, the report employs the classic Single-Factor Gaussian Copula Model under the Basel framework—essentially a Bernoulli Mixture Model—as the risk quantification model. Through 10,000 Monte Carlo simulations, the portfolio's Expected Loss (EL), Value at Risk (VaR), and Expected Shortfall (ES) were computed and extracted. The empirical results reveal the "tail amplification" mechanism driven by systematic macroeconomic shocks on extreme tail risk, and demonstrate the significant diversification benefits generated by merging portfolios across different regions and industries.

1. Data Description and Empirical Preprocessing

To construct a robust dataset suitable for simulation, a comprehensive data cleaning workflow was developed to address data quality issues within the raw data. This process systematically aligned and integrated portfolio holdings, entity characteristics, stock and index day returns, and 20-year historical default records. The cleaning process primarily resolved the following core data vulnerabilities:

1) Exposure at Default (EAD) Cleaning: Non-numeric text strings embedded within the EAD column were coerced into numeric values using `pd.to_numeric(errors = 'coerce')`, and invalid rows containing null exposures were dropped. Concurrently, highly inconsistent `booking_date` formats were standardized into uniform Format using a mixed date-parsing mechanism.

2) Loss Given Default (LGD) Cleaning: Highly fragmented industry nomenclature in the raw archives (e.g., *Ind. A*, *Industrya*) was strictly mapped to standardized "Industry A" and "Industry B" labels using a mapping dictionary combined with `.str.title()`. For missing LGD values among certain entities, the respective industry medians were calculated and imputed, preserving the underlying credit risk profile as closely as possible based on sector homogeneity.

3) Entity Code Cleaning: The Entity Code serves as the primary key connecting all data tables. Beyond standardizing global whitespace and capitalization, the program successfully captured and hardcoded fixes for critical missing hyphens in errant codes (e.g., rectifying *BWMAEKB* to *B-WMAEKB*). Furthermore, duplicate records were strictly removed when merging Portfolio 1 and Portfolio 2 to prevent double-counting exposures and avoid overestimating portfolio risk.

4) Defensive Filtering for Sample Size: When estimating the sensitivity to systematic factors (Beta), entities with extremely limited valid stock return data (less than 10 days) were identified. To prevent small samples from distorting linear OLS regression slopes and compromising the structural Gaussian factor framework, a defensive filtering mechanism was implemented to reject Beta estimation for these entities and exclude them, thereby ensuring the statistical significance of all remaining parameters.

Following this deep cleaning and parameter estimation, the scattered "messy data" was successfully consolidated into a high-quality Master Portfolio containing clean EAD, LGD, average Probability of Default (PD), and Beta parameters (saved in the "Data" folder).

2. Model Setup and Methodology Framework

To quantify the one-year credit risk of the aggregated portfolio, this study implements the Single-Factor Gaussian Copula Model, combined with the classic Threshold Model. From a statistical perspective, this framework is a standard Bernoulli Mixture Model.

2.1 Single-Factor Structure and Asset Value Evolution

The model assumes that the default event of the $N=56$ entities is driven by an unobservable, continuous latent variable (the critical variable X_i) representing asset return. X_i is driven by a systematic risk factor F (representing the global macroeconomic environment) and an idiosyncratic risk factor ε_i (representing firm-specific conditions). The mathematical formulation is $X_i = \beta_i * F + \eta_i * \varepsilon_i$, where both F and ε_i follow independent standard normal distributions. β_i is the sensitivity to the systematic factor, and the idiosyncratic weight η_i is strictly set to $\sqrt{(1 - \beta_i^2)}$ to maintain the standard normal distribution.

2.2 Default Trigger Mechanism and Parameter Mapping

An obligor i defaults if and only if its critical variable X_i falls below a specific default threshold C_i . Using the inverse of the standard normal cumulative distribution function, the historical average Probability of Default (PD) is mapped to the absolute threshold C_i . In this structure, conditional on the common systematic factor F , default events across obligors are independent. However, an adverse draw of F systematically lowers the asset values of all entities, perfectly simulating "dependence" and joint-default contagion effects.

2.3 Monte Carlo Simulation and Risk Measurement

Due to the high heterogeneity of EAD and LGD across assets, the overall loss distribution lacks an analytical solution. Therefore, Monte Carlo Simulation is used to

generate 10,000 future scenarios of systematic and idiosyncratic factors. Total portfolio losses are recorded in each run to construct the complete loss probability density distribution, *extracting Expected Loss (EL)*, *99% Value at Risk (VaR)*, and *99% Expected Shortfall (ES)*.

3. Empirical Results and Discussion

Through 10,000 Monte Carlo simulation paths based on the single-factor structural model, the portfolio's overall characteristics and quantitative credit risk metrics are computed as shown in Table 1 and Table 2 below.

Table 1: Portfolio Descriptive Statistics

Region / Industry	Number of Entities	Total EAD	Average LGD	Average PD
Industry A	28	631,058,707.46	0.4682	0.0196
Industry B	28	704,450,945.09	0.5001	0.0468
Total Portfolio	56	1,335,509,652.56	0.4842	0.0332

Table 2: Monte Carlo Simulation Core Risk Metrics

Risk Metric	Value	% of Total EAD
Total EAD	1,335,509,652.56	100.00%
Expected Loss (EL)	21,524,502.12	1.61%
99% VaR	249,445,444.52	18.68%
99% ES	298,671,602.35	22.36%

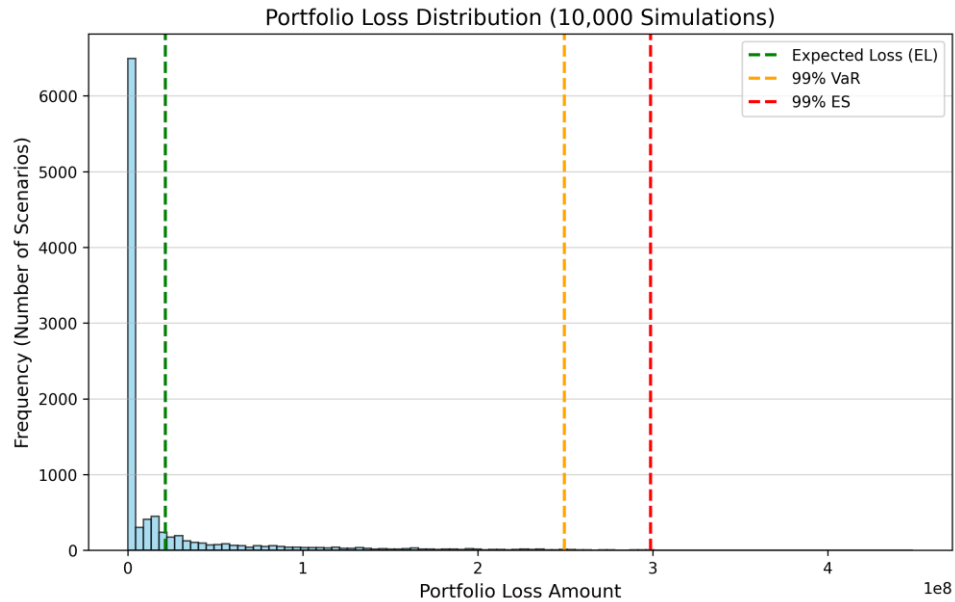


Figure 1: Simulated One-Year Loss Distribution (10,000 paths)

3.1 Economic Capital Implications of EL and UL

Firstly, the significant difference between the Expected Loss (EL, approximately 21.52 M) and VaR dictates distinct capital allocation strategies. EL represents a baseline operational cost. In contrast, UL—defined as the buffer between the 99% VaR and the EL (approximately 227.9 M)—poses the genuine threat to institutional solvency and therefore serves as the baseline requirement for core Economic Capital. This capital segregation mechanism ensures that the financial institution maintains adequate solvency even under extreme stress scenarios.

3.2 Tail Amplification and Fat-Tailed Distribution

Secondly, the vast nearly \$50M disparity between 99% *VaR* ($\sim 249.4M$) and 99% *Expected Shortfall* (*ES*, $\sim 298.6M$) highlights a highly right-skewed and "fat-tailed" credit loss distribution. This extreme "tail amplification" is a direct consequence of the model's factor structure: when the macroeconomic environment suffers severe distress, the high dependence leads to clustered, joint-default events.

3.3 Diversification Benefits of Portfolio Aggregation

Finally, the aggregated Master Portfolio demonstrates robust diversification benefits. Merging Portfolios 1 and 2 pools credit exposures across distinct geographic regions (Asia and South America) and separate industrial sectors (Industry A and Industry B). Since the firm-specific idiosyncratic risk factors (ε_i) are mutually independent, they effectively offset each other within the expanded multi-asset pool. Consequently, the consolidated portfolio exhibits significantly lower aggregated risk metrics than the simple sum of isolated sub-portfolios, validating the strategic value of cross-industry and cross-border diversification.

Conclusion

This empirical study systematically demonstrates governance strategies for handling "messy data" in modern commercial banking, and the profound value of employing a Single-Factor Gaussian Copula structural model for risk measurement. Rigorous mapping and imputation rules successfully eliminated cross-lending exposures and invalid parameters, ensuring data integrity. The 10,000-path simulation confirmed the severe fat-tailed nature of credit loss distributions and the impact of systemic risk contagion. This reminds risk managers that relying solely on the point-estimate VaR is insufficient; ES must be jointly evaluated to monitor deep tail threats, while actively pursuing portfolio diversification strategies to optimize economic capital consumption.

References

- Lütkebohmert-Holtz, E. & Shen, H. (2026). Introduction to Credit Risk and Applications in Python: Lecture Slides. University of Freiburg, Department of Quantitative Finance.
- Fitch Ratings. (2008). Basel II Correlation Values: An Empirical Analysis of EL, UL and the IRB Model. Credit Market Research, Special Report.